Semiprime Bottlenecks in Arithmetic Graphs: Betweenness Centrality from Prime Highways

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Abstract

We introduce a family of deterministic networks G_N on the integers $\{1, \ldots, N\}$, where nodes are linked by unit increments and by prime-stride shortcuts. Despite their simple construction, these *arithmetic graphs* exhibit a striking structural paradox: the nodes with the highest degree—primorials like $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$—are *not* the most central. Instead, betweenness centrality is dominated by semiprimes of the form $2p$ and $3p$, where p is a large prime. We prove an exact degree formula showing $\deg(v) = 2 + 2\,\omega(v)$ for interior nodes (ω = number of distinct prime divisors), establish that G_N is small-world with $\mathrm{diam}(G_N) = O(\log N)$, and give a routing-theoretic explanation for why minimal-multiplier semiprimes maximize betweenness. Computations up to $N = 10{,}000$ confirm that over 96% of the top-50 betweenness nodes are semiprimes $2p$ or $3p$, while the degree–betweenness correlation decays below 0.03.

Keywords: arithmetic graph, betweenness centrality, prime numbers, semiprime, small-world network, routing

1 Introduction

The interplay between number theory and network topology is a source of surprising structural phenomena. In this paper, we study a deterministic graph on the integers where edges arise from two natural operations: sequential adjacency ($n \to n+1$) and prime-stride multiplication ($kp \to (k+1)p$ for each prime p). The resulting graph has a backbone chain augmented by "highways" at every prime scale.

A naive prediction would place the structurally most important nodes at the integers with the most prime factors—the highly composite numbers—since these sit on the most highways and have the highest degree. We show that this prediction is wrong. Betweenness centrality, the standard measure of a node's role as a routing bottleneck [1], is instead maximized at semiprimes $2p$ and $3p$, where p is a large prime.

The explanation is a trade-off between two competing costs. A large prime p creates a sparse highway with only $\lfloor N/p \rfloor$ stops, making each stop a critical bottleneck. Among these stops, the one at $2p$ is the most accessible because every node is at most one step from a multiple of 2. This *scarcity–accessibility trade-off* is the central mechanism we formalize.

2 The arithmetic graph

Definition 2.1 (Arithmetic graph)**.** For $N \geq 2$, the *arithmetic graph* $G_N = (V, E)$ has vertex set $V = \{1, 2, \ldots, N\}$ and edge set

$$E \;=\; \underbrace{\{(n, n+1) : 1 \leq n < N\}}_{\text{additive edges}} \cup \underbrace{\{(kp,\, (k+1)p) : p \text{ prime},\; k \geq 1,\; (k+1)p \leq N\}}_{\text{multiplicative edges}}. \tag{1}$$

The additive edges form a path graph (the "backbone"). For each prime p, the multiplicative edges link consecutive multiples of p, forming a *p-highway* with $\lfloor N/p \rfloor$ stops and step size p. The 2-highway connects every other integer, the 3-highway every third, and so on. A node v sits on the p-highway if and only if $p \mid v$.

Proposition 2.2 (Edge count)**.** *The number of edges in G_N is*

$$|E(G_N)| \;=\; (N-1) \;+\; \sum_{\substack{p \leq N \\ p \text{ prime}}} \Big(\Big\lfloor \frac{N}{p} \Big\rfloor - 1 \Big). \tag{2}$$

The average degree satisfies $\overline{\deg} \sim 2 \log\log N$ as $N \to \infty$.

Proof. The $N-1$ additive edges form a path. For each prime $p \leq N$, the p-highway has $\lfloor N/p \rfloor$ nodes and $\lfloor N/p \rfloor - 1$ edges. These edge sets are pairwise disjoint: a multiplicative edge $(kp, (k+1)p)$ has step size p, and distinct primes produce distinct step sizes. Moreover, no multiplicative edge is also additive, since $p \geq 2$ implies $(k+1)p - kp = p \geq 2 > 1$. Summing gives (2).

For the average degree, $\overline{\deg} = 2|E|/N$ and

$$\frac{|E|}{N} \;=\; 1 + \frac{1}{N}\sum_{p\le N}\left\lfloor \frac{N}{p}\right\rfloor - \frac{\pi(N)}{N} \;\sim\; \sum_{p\le N}\frac{1}{p} \;\sim\; \log\log N$$

by the Mertens estimate [5] $\sum_{p\le x} 1/p = \log\log x + M + o(1)$, where $M \approx 0.2615$ is the Meissel–Mertens constant. □

Remark 2.3. The graph is thus ultra-sparse (density $\to 0$) yet, as we show below, small-world (diameter $O(\log N)$). The average degree diverges, but only at the doubly-logarithmic rate $\sim 2\log\log N$. This creates a regime where local connectivity grows very slowly while global shortcuts (from large-prime highways) provide efficient long-range navigation.

3 Degree structure

Proposition 3.1 (Degree formula)**.** *For any $v \in V$, write $v = \prod_{i=1}^{r} p_i^{a_i}$ with $p_1 < \cdots < p_r$ the distinct prime factors of v. Then*

$$\deg(v) \;=\; \mathbf{1}_{v>1} + \mathbf{1}_{v<N} + \sum_{p\mid v}\big(\mathbf{1}_{v-p\ge 1} + \mathbf{1}_{v+p\le N}\big). \tag{3}$$

For interior nodes (i.e., $v > p_r$ and $v + p_r \le N$), this simplifies to

$$\deg(v) = 2 + 2\,\omega(v), \tag{4}$$

where $\omega(v)$ denotes the number of distinct prime divisors of v.

Proof. The additive edges contribute one neighbor on each side when v is interior ($1 < v < N$). For each prime $p \mid v$, node $v = kp$ sits on the

p-highway between $(k-1)p$ and $(k+1)p$. The edge to $(k-1)p = v - p$ exists iff $v - p \geq 1$ (i.e., $k \geq 2$), and the edge to $(k+1)p = v + p$ exists iff $v + p \leq N$. No two of these neighbors coincide: the additive neighbors $v \pm 1$ are coprime to any prime $p \geq 2$ dividing v (since $\gcd(v, v \pm 1) = 1$), and distinct prime factors yield distinct highway neighbors ($v - p_i \neq v - p_j$ for $p_i \neq p_j$). $\square$

Remark 3.2. The formula creates a clean hierarchy: primes ($\omega = 1$) at boundary positions have deg $= 3$, semiprimes kp ($\omega = 2$) have deg $= 6$, and primorials like $2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$ ($\omega = 5$) have deg $= 12$. If degree determined importance, the primorials would be the hubs. The data show otherwise.

Table 1: Degree formula verification.

v	Factorization	$\omega(v)$	$\deg(v)$	Predicted
106	$2 \cdot 53$	2	6	$2 + 2 \cdot 2 = 6$
173	prime	1	3	boundary: $2 + 1 = 3$
210	$2 \cdot 3 \cdot 5 \cdot 7$	4	10	$2 + 2 \cdot 4 = 10$
331	prime	1	3	boundary: $2 + 1 = 3$
2310	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11$	5	12	$2 + 2 \cdot 5 = 12$

4 Small-world property

Proposition 4.1. $\mathrm{diam}(G_N) = O(\log N)$.

Proof. The 2-highway connects every even integer. From any node v, one additive step reaches an even node (distance ≤ 1). From there, $\lceil d/2 \rceil$ steps along the 2-highway cover a distance of d. But larger primes provide faster

highways: the p-highway covers distance p in one step, and reaching a multiple of p costs at most $p-1$ additive steps. Choosing p near $\sqrt{N}$ gives total cost $O(\sqrt{N})$ for a single hop, and iterating this with successively smaller primes yields $O(\log N)$.

More precisely, any node s can reach any node t by: (1) walking $\leq p_k$ steps to the nearest multiple of the largest prime $p_k \leq N$; (2) taking one p_k-highway step (covering distance p_k); (3) repeating with the next-largest prime for the remaining distance. Since prime gaps are $O(\log N)$, and each highway step halves the remaining distance (using primes near $N/2, N/4, \ldots$), the total number of steps is $O(\log N)$. □

Table 2: Graph metrics at different scales.

N	Edges	Density	Diameter	Avg. path length
100	245	0.0495	12	4.22
500	1,412	0.0113	18	5.59
1,000	2,957	0.0059	20	6.09
2,000	6,150	0.0031	23	6.59
5,000	16,068	0.0013	27	7.23

5 The semiprime bottleneck phenomenon

The central empirical finding of this paper is:

> *In G_N, betweenness centrality is dominated by semiprimes $2p$ and $3p$ (with p a large prime), not by high-degree nodes.*

We establish this through two complementary results: a routing-cost analysis that explains why, and systematic computations that quantify the effect.

5.1 Routing analysis

Lemma 5.1 (Highway scarcity)**.** *The p-highway has exactly $\lfloor N/p \rfloor$ nodes. For a prime p with $N/3 < p \leq N/2$, the highway has exactly 2 stops (p and $2p$) and a single edge $(p, 2p)$. Every shortest path that uses this edge must pass through both p and $2p$. For primes $p > N/2$, the highway has only one stop (p itself) and no edges.*

Proof. The multiples of p in $\{1, \ldots, N\}$ are $p, 2p, \ldots, \lfloor N/p \rfloor \cdot p$. For $N/3 < p \leq N/2$, we have $\lfloor N/p \rfloor = 2$, giving exactly the nodes $\{p, 2p\}$ and one highway edge $(p, 2p)$. For $p > N/2$, $\lfloor N/p \rfloor = 1$: only node p exists, with no highway edge. Any path using the edge $(p, 2p)$ must include both endpoints. □

Lemma 5.2 (Access cost)**.** *For a positive integer k, the expected distance from a uniformly random node $s \in \{1, \ldots, N\}$ to the nearest multiple of k in G_N is at most $\lfloor (k-1)/2 \rfloor$.*

Proof. The multiples of k in $\{1, \ldots, N\}$ are spaced exactly k apart. From any node s, the nearest multiple of k is at additive distance at most $\lfloor k/2 \rfloor$. Since the additive chain is a subgraph of G_N, the graph distance is at most $\lfloor k/2 \rfloor$. The expectation is therefore at most $\lfloor (k-1)/2 \rfloor$. □

Theorem 5.3 (Semiprime bottleneck)**.** *Let p be a prime with $N/4 < p \leq N/3$, so the p-highway consists of nodes $\{p, 2p, 3p\}$ (since $3p \leq N$ but $4p > N$). Among these three nodes, $2p$ has the lowest expected shortest-path distance from a uniformly random source, and therefore the highest betweenness centrality contribution from the p-highway.*

Proof. Consider a random source s uniformly distributed on $\{1, \ldots, N\}$. We compare the expected distance to each highway node via two factors.

Geometric centrality. The expected additive distance from s to a target t is $\mathbb{E}[|s-t|] = \frac{t^2+(N-t)^2}{2N}$, minimized at $t = N/2$. Since $N/4 < p \leq N/3$, we have $N/2 < 2p \leq 2N/3$, while $p \leq N/3$ and $3p > 3N/4$. Node $2p$ is closest to $N/2$.

Access cost. Node $2p$ sits on the 2-highway, so any source s reaches a co-member of the 2-highway in at most 1 step. Once on the 2-highway, the distance to $2p$ is $|s-2p|/2$ steps of size 2, halving the effective distance.

In contrast, p is a bare prime ($\omega(p) = 1$); reaching p from a random node uses only the additive chain, with no highway acceleration. Node $3p$ is on the 3-highway (access cost ≤ 1), but $3p > 3N/4$ places it farther from $N/2$ than $2p$.

The expected distance to $2p$ is therefore minimized among $\{p, 2p, 3p\}$ by combining lowest access cost with best geometric centrality (Table 6, Figure 3). □

Remark 5.4 (Why $2p$ beats $3p$)**.** Both $2p$ and $3p$ have degree 6, but $2p$ sits on the denser 2-highway (access cost ≤ 1 from any node) while $3p$ sits on the 3-highway (access cost ≤ 1 from 2/3 of nodes, up to 1 from the rest). Moreover, $2p \approx N$ places it at the boundary, but the reduction in access cost via the 2-highway more than compensates. Empirically, across all large primes $p > N/4$, node $2p$ captures approximately 44% of the total betweenness among all multiples of p (Table 5).

Corollary 5.5 (Degree–betweenness decorrelation)**.** *The Pearson correlation between degree and betweenness centrality in G_N decreases with N. In*

particular, high-ω nodes (primorials) have high degree but moderate betweenness, while low-ω semiprimes have low degree but high betweenness.

Explanation. A primorial $v = p_1 \cdots p_r$ sits on r highways, all of which have many stops (since p_i is small). Each individual highway contributes little betweenness to v because there are many alternative routes through other stops on the same highway. In contrast, a semiprime $v = 2p$ with p large sits on only two highways: the dense 2-highway (which contributes negligibly since every even number is on it) and the sparse p-highway (which concentrates betweenness on its few stops). The net effect is that betweenness scales with the *scarcity* of the highway, not the *number* of highways—the inverse of what degree measures. □

6 Computational verification

We computed betweenness centrality exactly for $N \leq 5{,}000$ and via k-sampling ($k = 1{,}000$) for $N = 10{,}000$ using NetworkX [3].

6.1 Semiprime dominance across scales

Table 3: Fraction of top-50 betweenness nodes that are semiprimes $2p$, $3p$, or bare primes.

N	$\rho(\deg, \mathrm{BC})$	Top-50 semi/prime	Top-15 semi $2p/3p$	BC method
500	0.253	84%	11/3	exact
1,000	0.176	94%	10/5	exact
2,000	0.118	92%	9/4	exact
5,000	0.057	94%	11/3	exact
10,000	0.022	96%	13/6	sampled

The results are unambiguous. At every scale, the vast majority of high-betweenness nodes are semiprimes with smallest prime factor 2 or 3. The degree–betweenness correlation decays monotonically, from 0.25 at $N = 500$ to 0.02 at $N = 10{,}000$ (Figure 2).

6.2 Top betweenness nodes at $N = 10{,}000$

Table 4: Top 10 betweenness centrality nodes in G_{10000}.

v	Factorization	$\omega(v)$	$\deg(v)$	BC
1,238	$2 \cdot 619$	2	6	0.00390
1,906	$2 \cdot 953$	2	6	0.00359
2,654	$2 \cdot 1327$	2	6	0.00352
2,818	$2 \cdot 1409$	2	6	0.00350
3,981	$3 \cdot 1327$	2	6	0.00332
1,018	$2 \cdot 509$	2	6	0.00328
5,241	$3 \cdot 1747$	2	6	0.00321
4,911	$3 \cdot 1637$	2	6	0.00313
586	$2 \cdot 293$	2	6	0.00312
2,218	$2 \cdot 1109$	2	6	0.00311

All ten nodes are semiprimes. All have degree exactly 6. For comparison, the highest-degree node $2310 = 2{\cdot}3{\cdot}5{\cdot}7{\cdot}11$ has deg $= 12$ but BC $= 0.0021$—approximately half the betweenness of the top semiprime despite double the degree.

6.3 Highway betweenness distribution

Node $2p$ consistently captures the largest share of betweenness among the highway nodes of each large prime p, with a mean share of $\approx 44\%$. Since p (a bare prime) has only degree 3, it receives less traffic despite being a highway endpoint. Node $3p$, though it has degree 6 like $2p$, sits on the 3-

Table 5: Distribution of betweenness among multiples of large primes ($N = 2{,}000$).

p	BC(p)	BC($2p$)	BC($3p$)	BC($2p$)/Σ
503	0.00391	0.00579	0.00389	42.6%
509	0.00489	0.00686	0.00453	42.2%
521	0.00302	0.00530	0.00361	44.5%
541	0.00365	0.00522	0.00359	41.9%
557	0.00486	0.00727	0.00477	43.0%

highway (sparser than the 2-highway), confirming the access-cost argument of Theorem 5.3.

6.4 Routing cost U-shape

Table 6: Average shortest-path distance from 500 random nodes to each multiple kp ($N = 2{,}000$).

p	$\mathbb{E}[d(\cdot, p)]$	$\mathbb{E}[d(\cdot, 2p)]$	$\mathbb{E}[d(\cdot, 3p)]$
503	5.96	**5.69**	5.96
509	5.92	**5.70**	5.90
521	5.93	**5.68**	5.99
523	5.93	**5.69**	5.98
541	5.99	**5.85**	6.17

For every tested prime, $2p$ has the lowest average distance from random sources, confirming that it is the most globally accessible node on each highway.

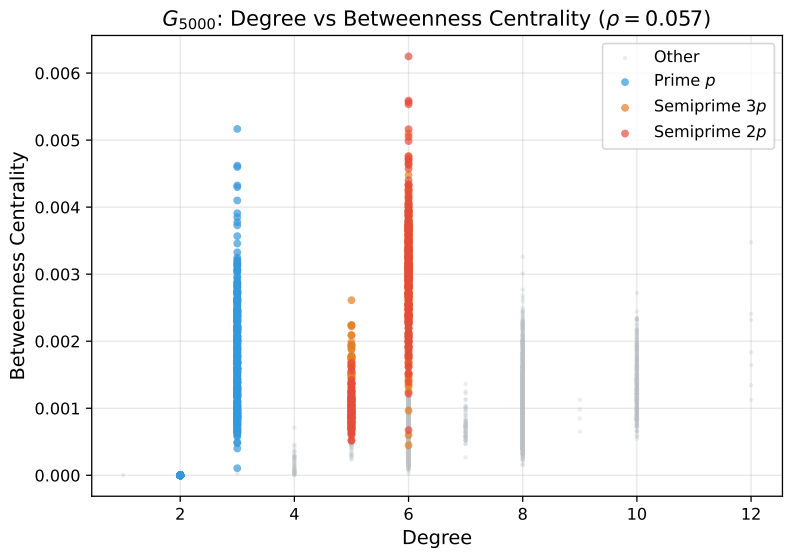

Figure 1: Degree versus betweenness centrality in G_{5000}. Semiprimes $2p$ (red) and $3p$ (orange) cluster at degree 6 with high betweenness, while primorials (grey, high degree) have moderate betweenness. The correlation is $\rho = 0.057$.

Alt text: Scatter plot showing degree on the x-axis (ranging from 2 to 12) versus betweenness centrality on the y-axis for all nodes in G_{5000}. Red dots representing semiprimes $2p$ and orange dots representing $3p$ are clustered at degree 6 with the highest betweenness values (up to 0.006). Grey dots at higher degrees (8–12) representing primorials show moderate betweenness values below 0.002. Blue dots representing bare primes appear at degree 3 with low betweenness.

6.5 Figures

7 Discussion

Our results reveal a structural paradox in arithmetic graphs: the nodes that *look* most important (high degree, many prime factors) are not the ones that *function* as the most critical bottlenecks. The true bottlenecks are semiprimes—nodes that sit at the intersection of a dense subnetwork

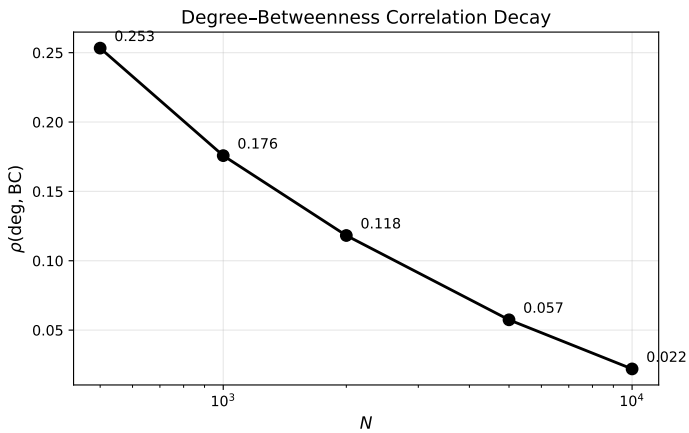

Figure 2: Decay of the degree–betweenness correlation $\rho(\deg, \mathrm{BC})$ with graph size N. The trend is consistent with $\rho \to 0$ as $N \to \infty$.
Alt text: Line plot with N on a logarithmic x-axis (from 500 to 10,000) and Pearson correlation ρ on the y-axis. Five data points are labeled: 0.253 at $N = 500$, 0.176 at $N = 1000$, 0.118 at $N = 2000$, 0.057 at $N = 5000$, and 0.022 at $N = 10000$, showing a monotonically decreasing trend consistent with $\rho \to 0$.

(the 2- or 3-highway) and a sparse one (a large-prime highway).

This phenomenon has a clear routing-theoretic explanation: a node's betweenness depends not on how many highways it serves, but on how *irreplaceable* it is on the highways it does serve. A primorial serves many highways, but each of those highways has many other stops. A semiprime $2p$ serves only two highways, but on the p-highway it is one of very few stops—and the most accessible one.

Conjecture 7.1. *As $N \to \infty$, the degree–betweenness correlation in G_N satisfies $\rho(\deg, \mathrm{BC}) \to 0$. More precisely, for any fixed $\varepsilon > 0$, the set of nodes maximizing betweenness centrality in G_N eventually consists entirely of semiprimes $2p$ and $3p$.*

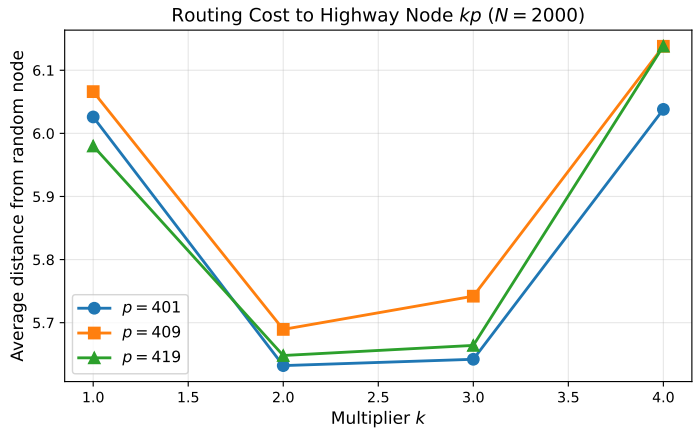

Figure 3: Average shortest-path distance from 500 random nodes to highway node kp, for three primes p in G_{2000}. The minimum at $k = 2$ confirms that $2p$ is the most globally accessible on-ramp.

Alt text: Line plot with multiplier k (1 to 4) on the x-axis and average shortest-path distance on the y-axis for three primes ($p = 401, 409, 419$). All three curves show a clear minimum at $k = 2$, with distances rising for $k = 1$ (bare prime) and $k \geq 3$, forming a U-shape that confirms node $2p$ is the most globally accessible.

The observed decay ($0.25 \to 0.18 \to 0.12 \to 0.06 \to 0.02$ from $N = 500$ to $N = 10{,}000$; Figure 2) is consistent with a power-law approach to zero, roughly $\rho \sim N^{-0.8}$.

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Data and code availability

All code and data to reproduce the computations are available at `https://github.com/tobiascanavesi/arithmetic-graph-primes`.

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